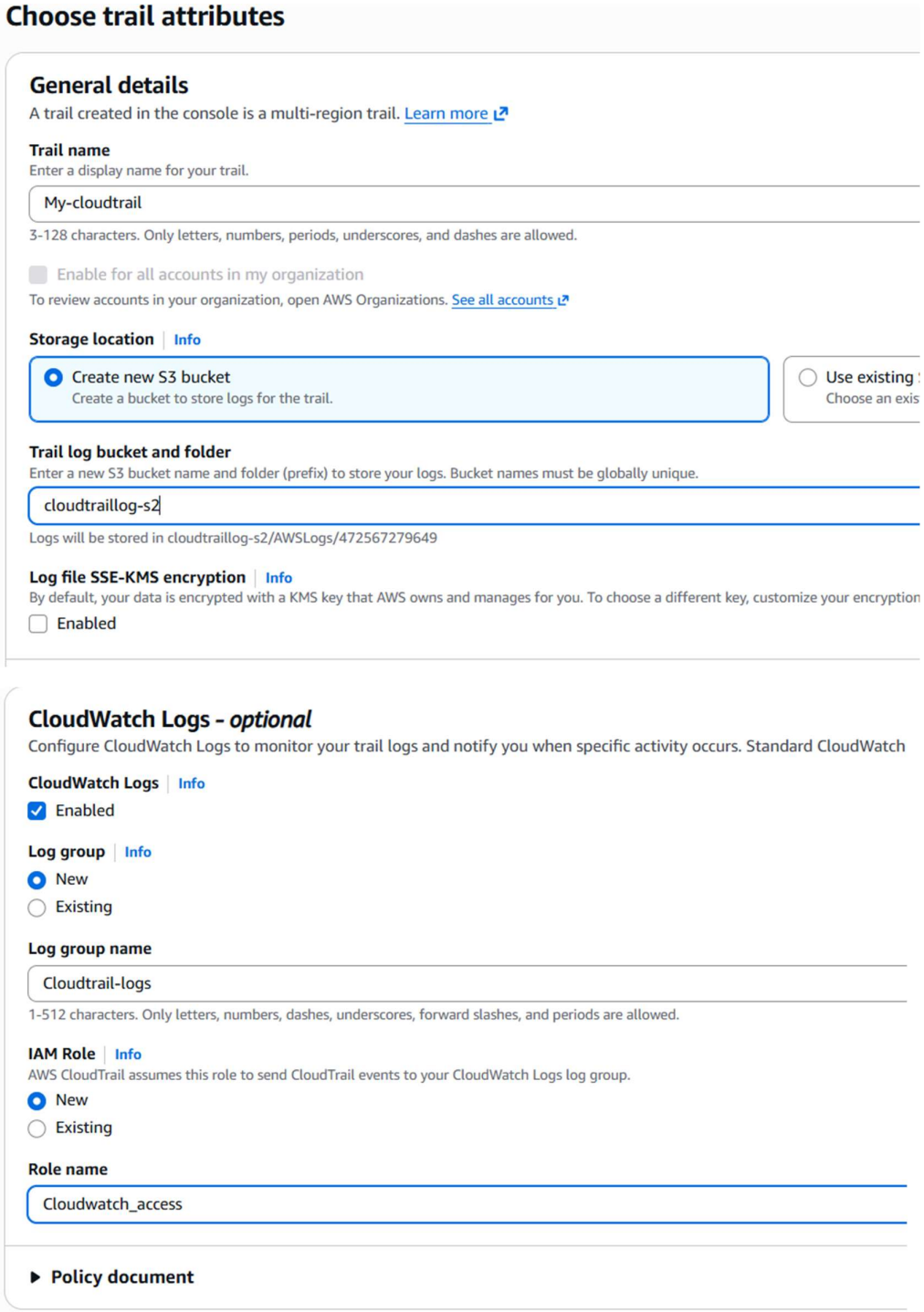
Cloudtrail & Cloudwatch Task

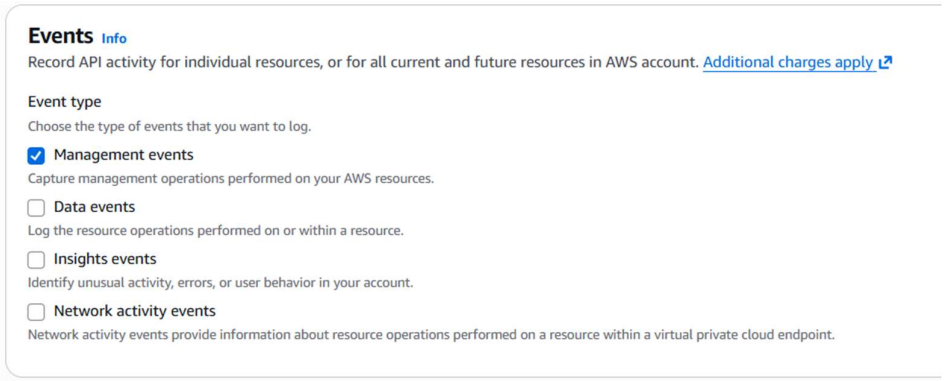
1. Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

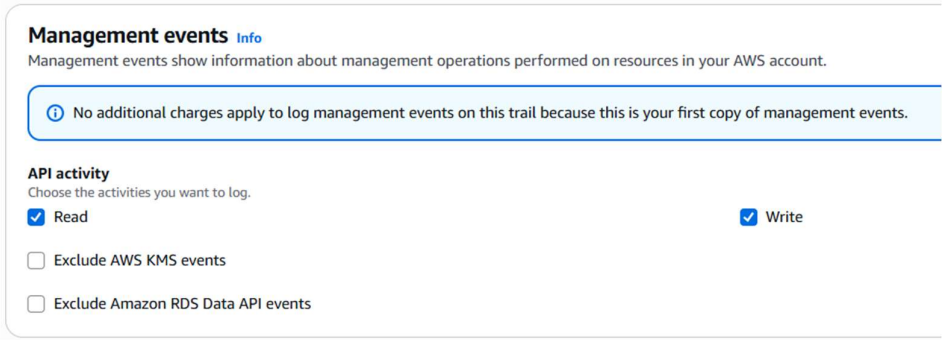
Step 1:- Go to cloudtrail services and create a new trail configure the s3, cloudwatch in it.

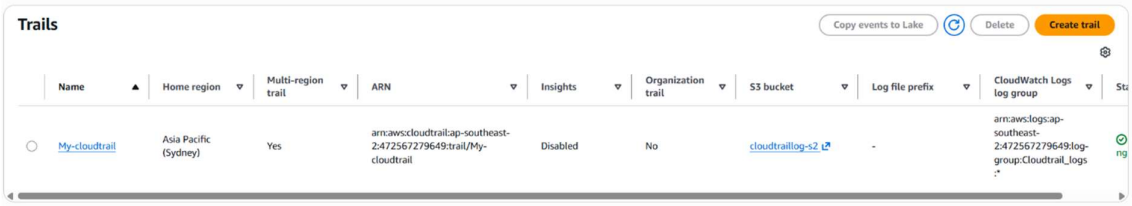
Step 2:- Then at page 2 select Management Events and give write & read permission

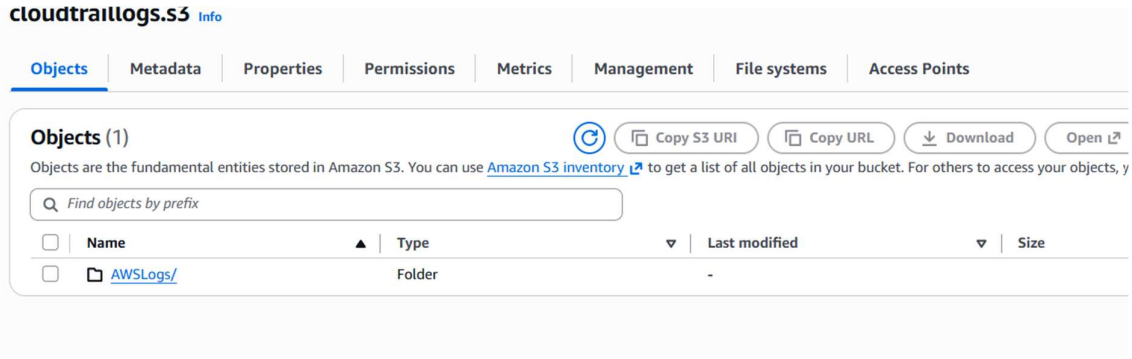
Step 3:- After the setup we can see the activity logs in both the services.

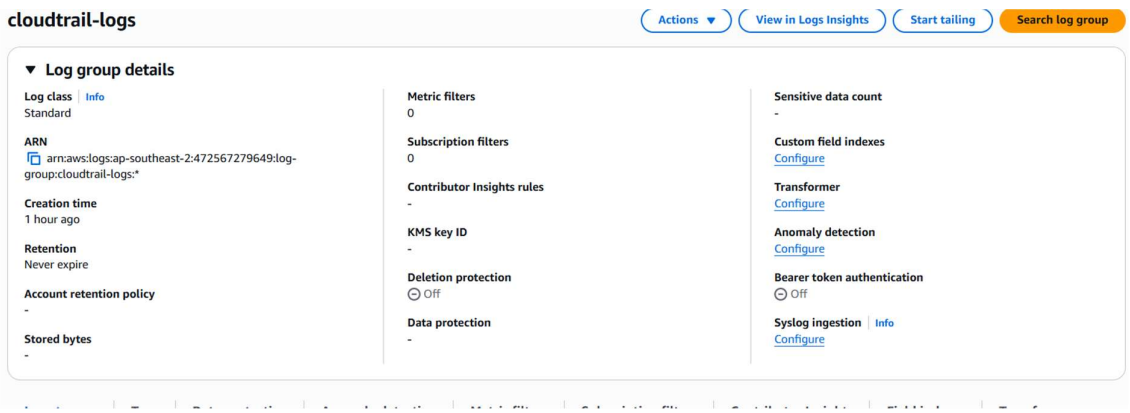










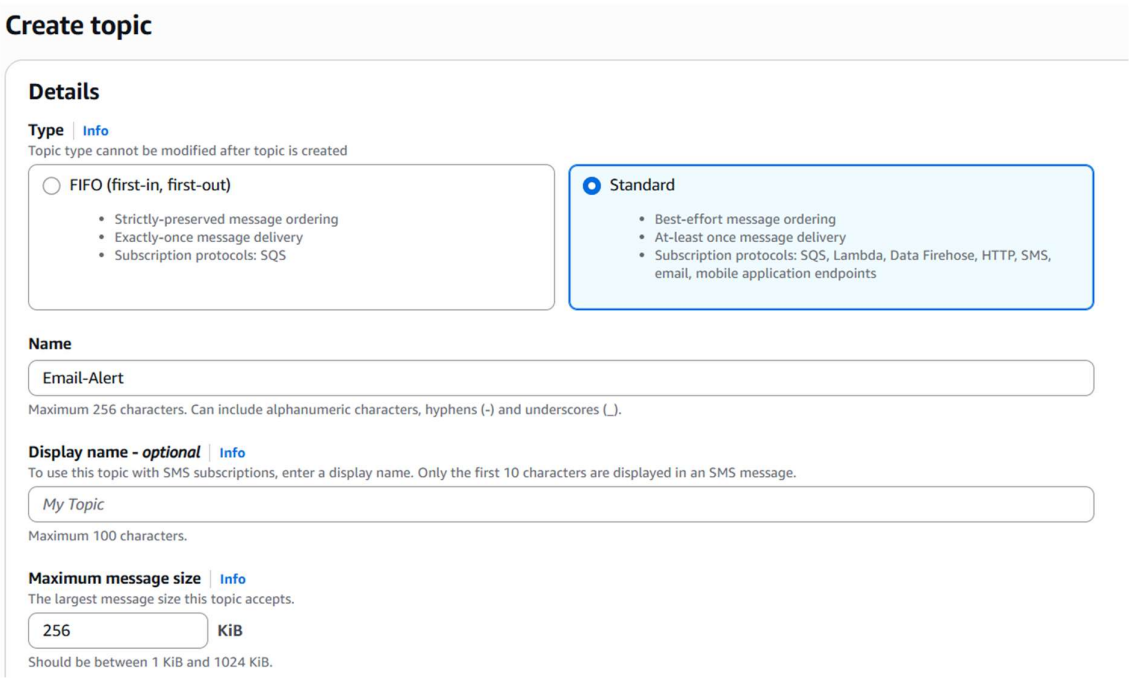


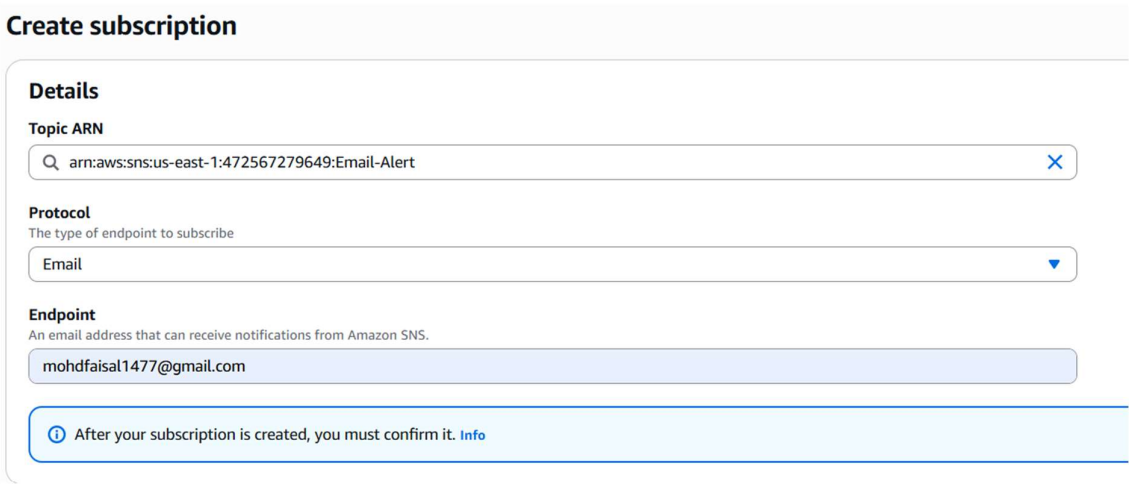
2. Enable SNS for CloudTrail to send alerts via email.

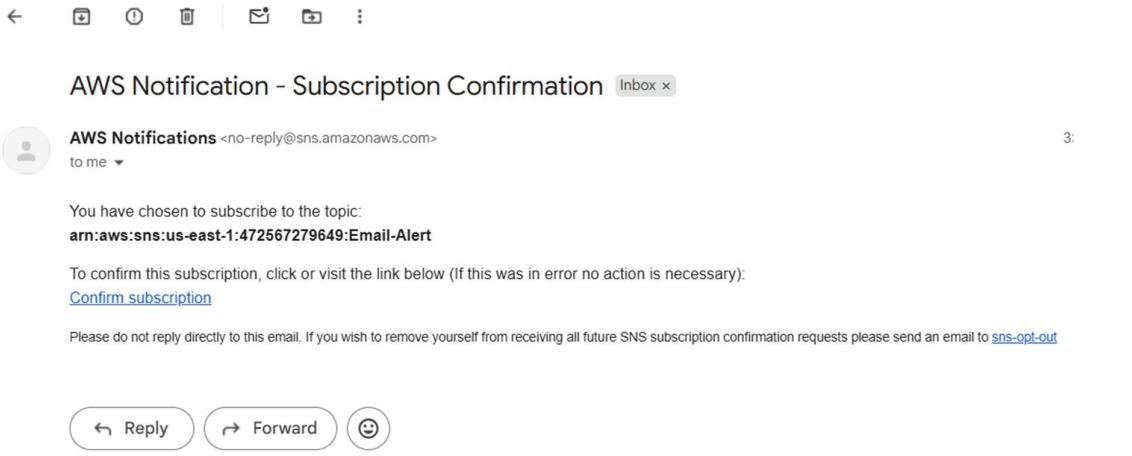
Step 1:- Open SNS section, create a new topic give a topic name.

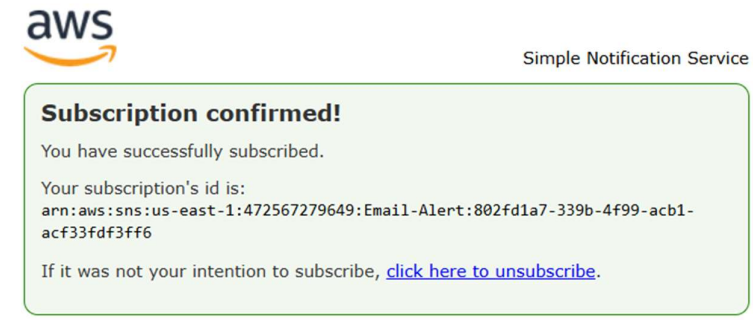
Step 2:- after creating the SNS go to its subscription, then configure the email protocol.

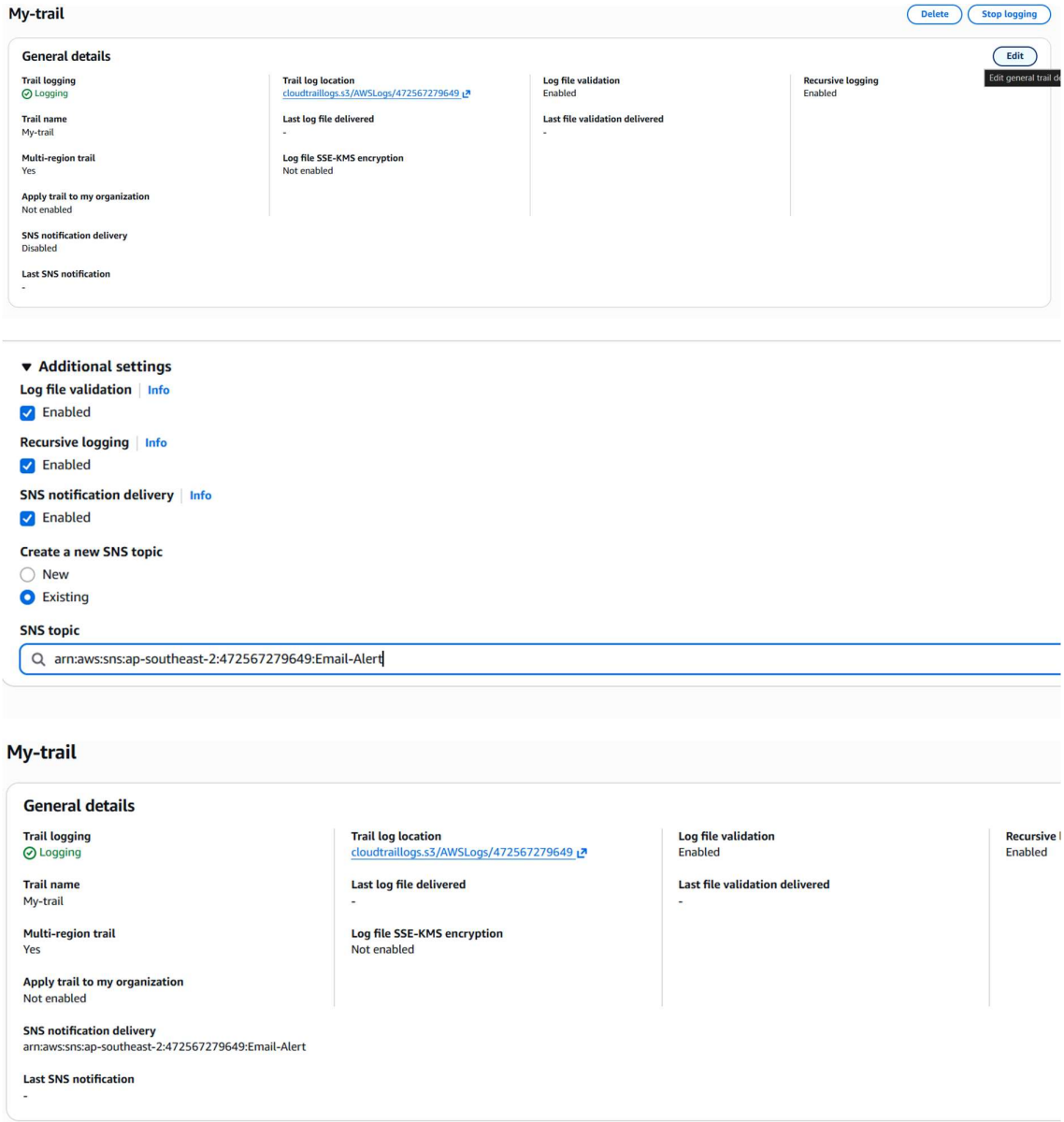
Step 3:- Now go to Cloudtrail and assign the SNS to Cloudtrail.









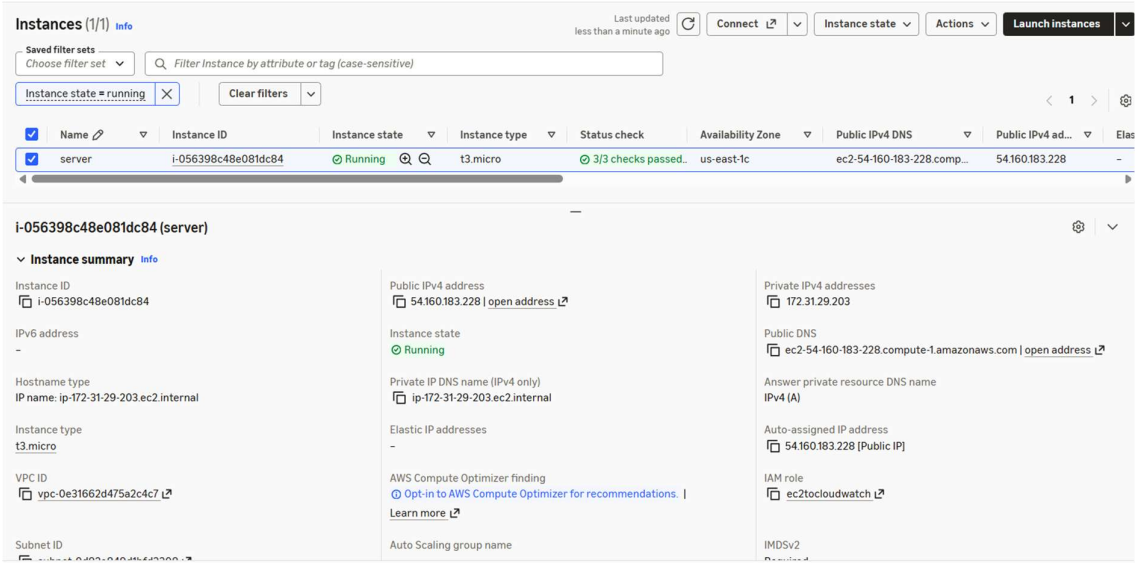


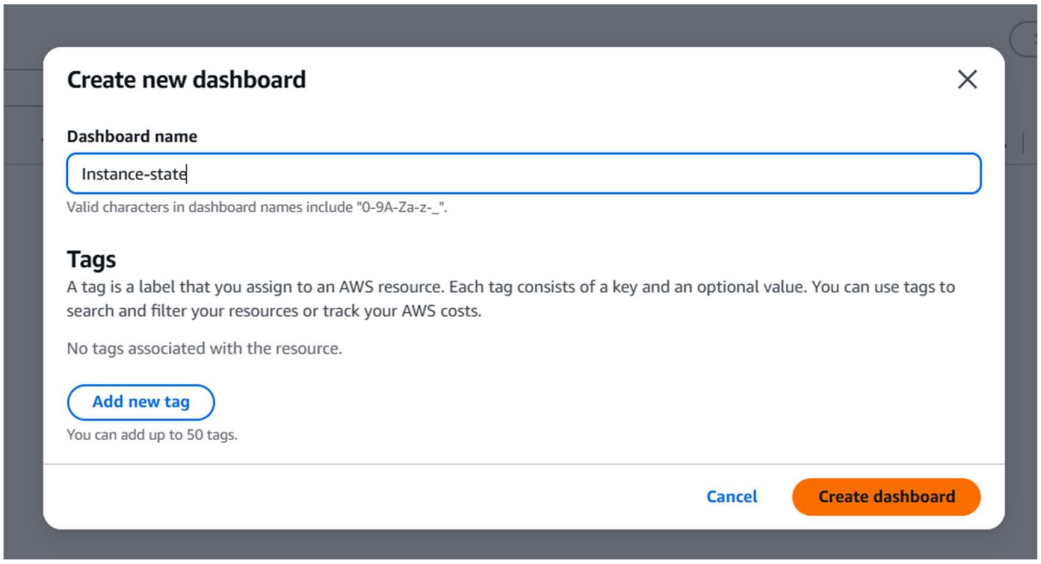
3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

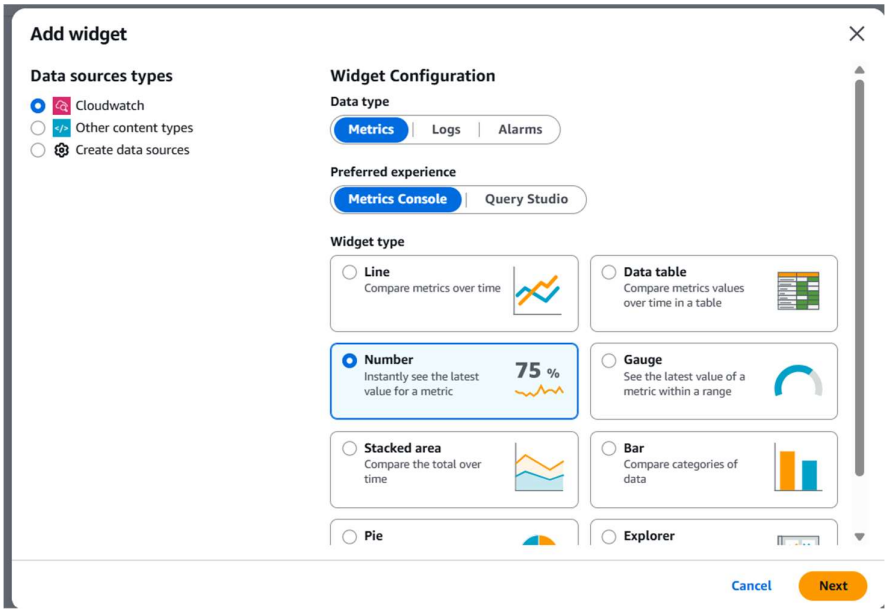
Step 1:- First launch a instance then go to cloudwatch services.

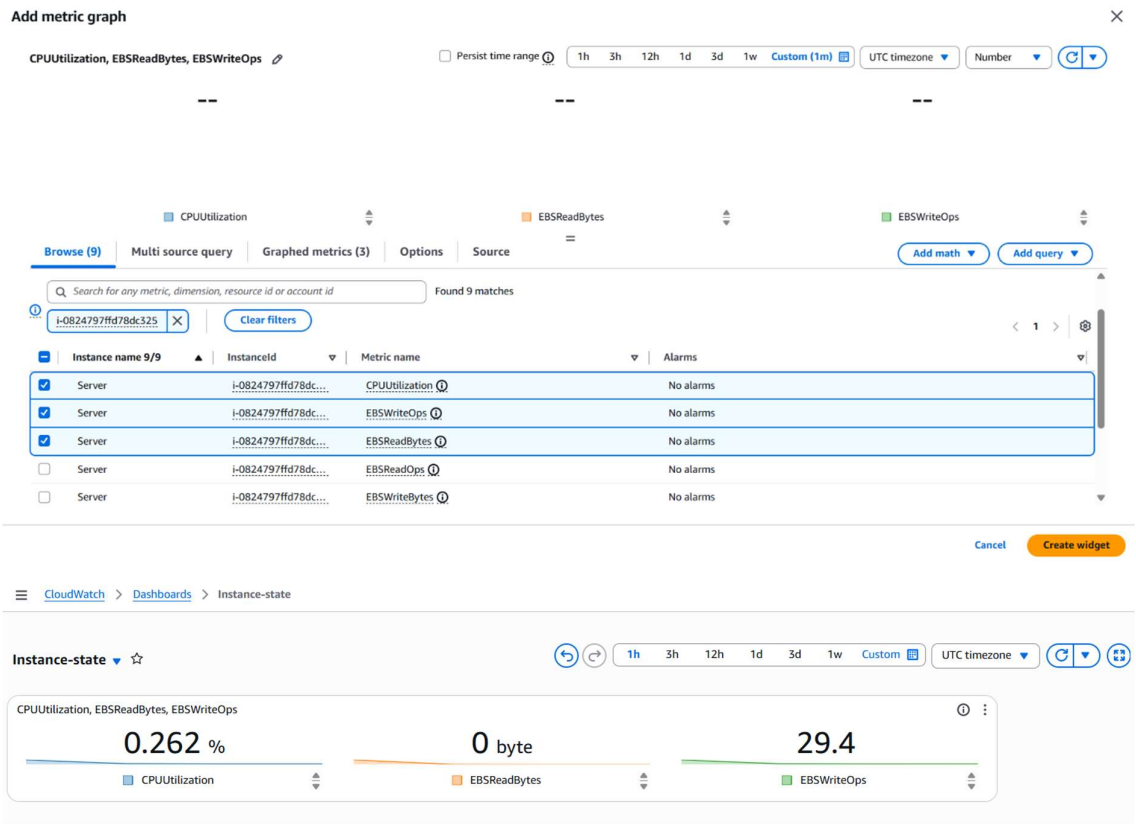
Step 2:- Go to dashboards create a new dashboard and add some widget by filtering the instance id. Save it

Step 3:- Now we can see the CPU utilization and other metrics.









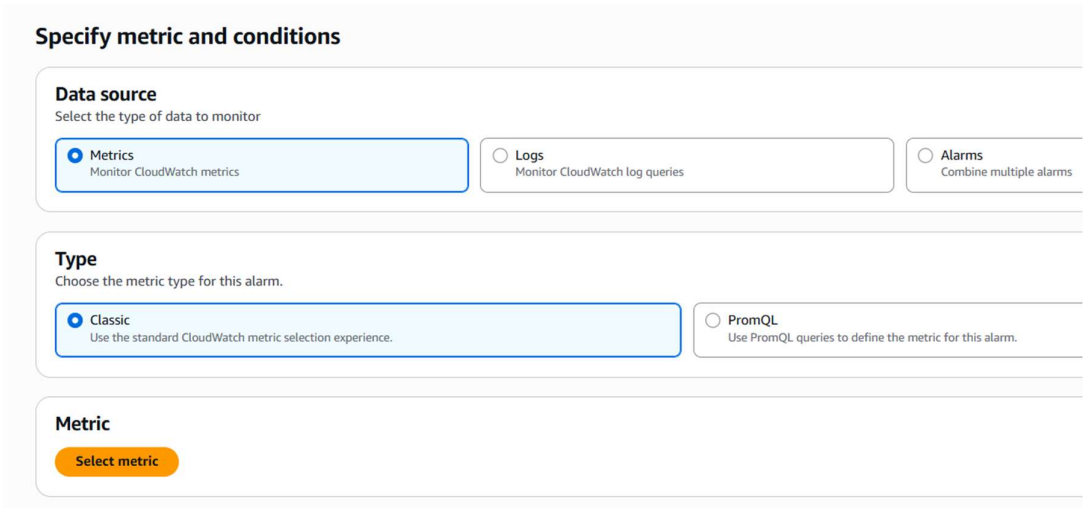
4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

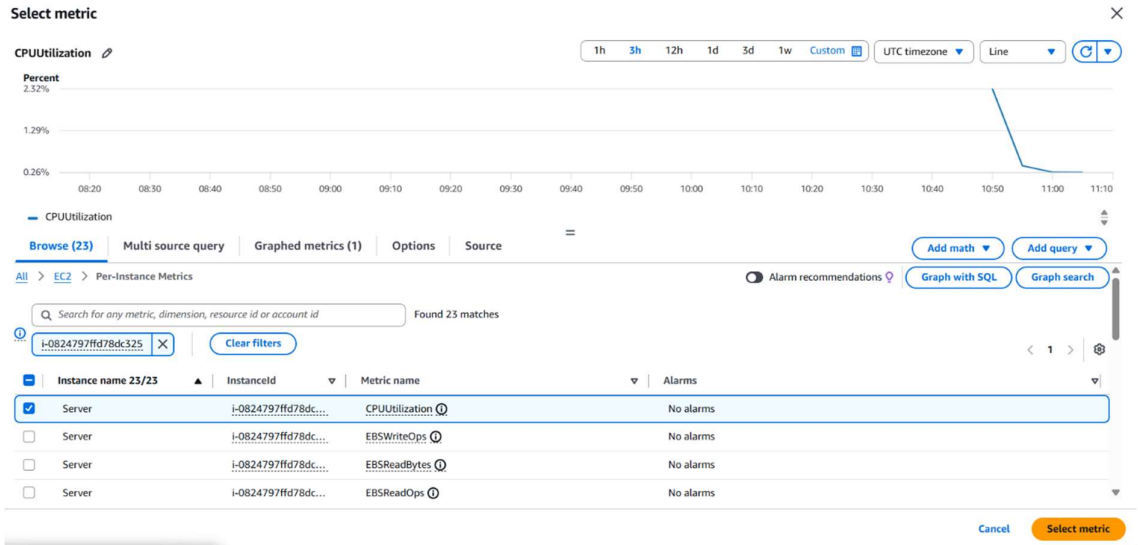
Step 1:- We need SNS for this. Then in Cloudwatch select alarms.

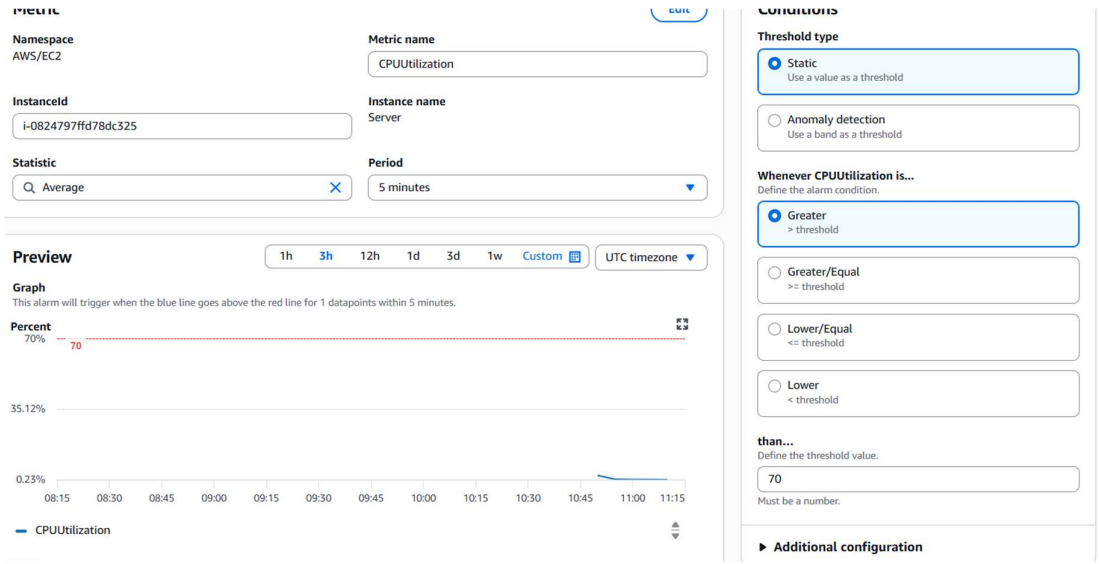
Step 2:- Then select the metrics to which we want to assign the alarm, then scroll to add condition as if greater than 70.

Step 3:- After that set the alarm state to In alarm and select the SNS.

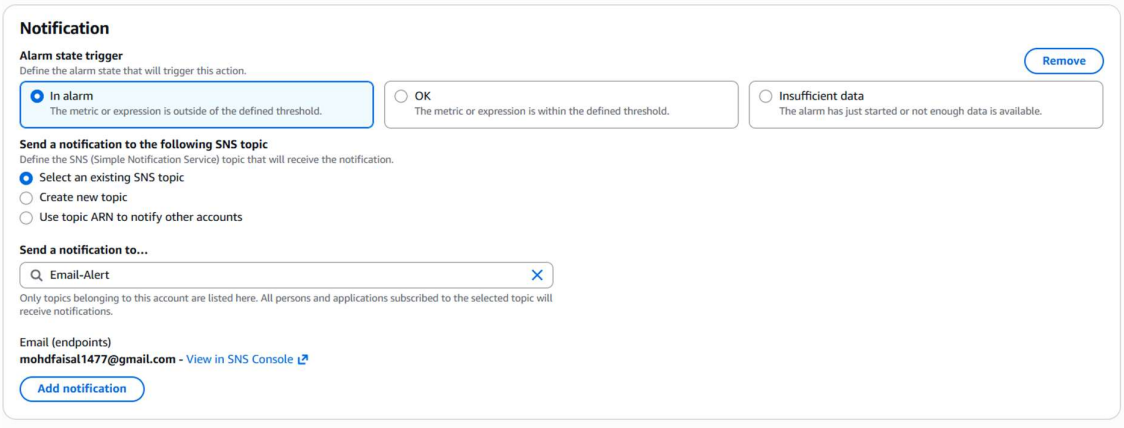
Step 4:- Now name the alarm.

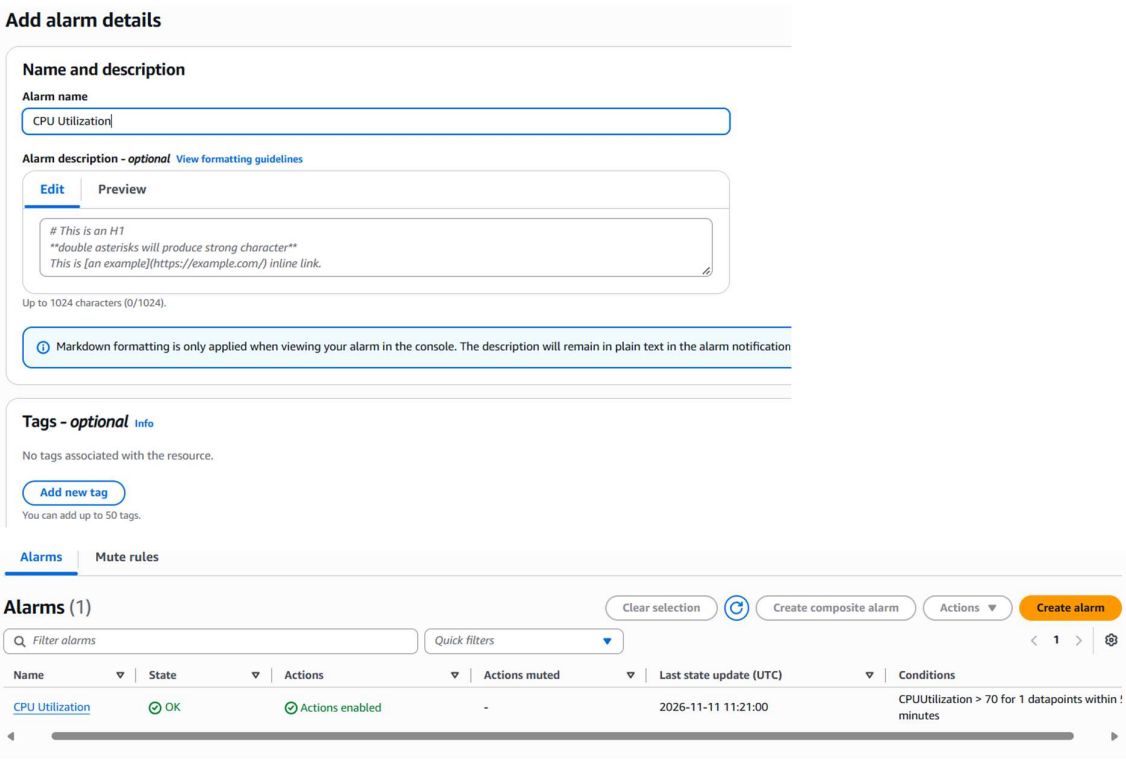






Configure actions





5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

Step 1:- First create a role with permission cloudwatchagentserver-policy.

Step 2:- Assign this role to a instance. Then open the instance and install java, tomcat packages.

Step 3:- Now create a bash script to monitor the tomcat status and send it to cloudwatch.

Step 4:- Manually start the tomcat package then run the bashscript.

Step 5:- Now in cloudwatch dashboard when we click on new widget we can see new metric tomcat is added.

Step 6:- click the metric and add it to dashboard now edit the time to one minute.

Step 7:- To continuously monitor the service add the script to cronjobs.

Step 8:- Go to alarms to add new alarm set the condition such if the metric value is 1 status is ok.

Step 9:- If the status is 0 then the service file is not working. Create alarms for both conditions.

ec2tocloudwatch Info

Delete

Allows EC2 instances to call AWS services on your behalf.

Summary

Edit

Creation date
November 11, 2026, 18:05 (UTC+05:30)

ARN
arn:aws:iam::472567279649:role/ec2tocloudwatch

Instance profile ARN
arn:aws:iam::472567279649:instance-profile/ec2tocloudwatch

Last activity
-

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (1) Info

You can attach up to 20 managed policies.

Simulate | Remove | Add permissions

Filter by Type: All types

Search

Policy name	Type	Attached entities
CloudWatchAgentServerPolicy	AWS managed	1

Modify IAM role Info

Attach an IAM role to your instance.

Instance ID
i-0824797ffd78dc325 (Server)

IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

ec2tocloudwatch | Create new IAM role

Cancel | Update IAM role

Instances (1/1) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set | Filter Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ad...	Elas
server	i-056398c48e081dc84	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-160-183-228.comp...	54.160.183.228	-

i-056398c48e081dc84 (server)

Instance summary Info

Instance ID: i-056398c48e081dc84	Public IPv4 address: 54.160.183.228 | open address	Private IPv4 addresses: 172.31.29.203
IPv6 address: -	Instance state: Running	Public DNS: ec2-54-160-183-228.compute-1.amazonaws.com | open address
Hostname type: IP name: ip-172-31-29-203.ec2.internal	Private IP DNS name (IPv4 only): ip-172-31-29-203.ec2.internal	Answer private resource DNS name: IPv4 (A)
Instance type: t3.micro	Elastic IP addresses: -	Auto-assigned IP address: 54.160.183.228 [Public IP]
VPC ID: vpc-0e31662d475a2c4c7	AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. | Learn more	IAM role: ec2tocloudwatch
Subnet ID	Auto Scaling group name	IMDSv2

```
[root@ip-172-31-29-203 ~]# yum install java-21 tomcat10 -y
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Thu Nov 12 06:38:55 2026.
Dependencies resolved.
================================================================================
 Package                          Architecture          Version
================================================================================
Installing:
 java-21-amazon-corretto          x86_64                1:21.0.13.0.11-1.amzn2023.1
 tomcat10                         noarch                1:10.1.60-1.amzn2023.0.1
Installing dependencies:
 alsa-lib                         x86_64                1.2.7.2-1.amzn2023.0.4
 apr                              x86_64                1.7.5-1.amzn2023.0.4
 cairo                            x86_64                1.18.0-4.amzn2023.0.3
 dejavu-sans-fonts                noarch                2.37-16.amzn2023.0.2
 dejavu-sans-mono-fonts           noarch                2.37-16.amzn2023.0.2
 dejavu-serif-fonts               noarch                2.37-16.amzn2023.0.2
 ecj                              noarch                1:4.23-1.amzn2023.0.4
 fontconfig                       x86_64                2.13.94-2.amzn2023.0.2
 fonts-filesystem                 noarch                1:2.0.5-12.amzn2023.0.2
 freetype                         x86_64                2.13.2-5.amzn2023.0.2
 giflib                           x86_64                5.2.1-9.amzn2023.0.4
 google-noto-fonts-common         noarch                20240401-1.amzn2023.0.2
 google-noto-sans-vf-fonts        noarch                20240401-1.amzn2023.0.2
 graphite2                        x86_64                1.3.14-7.amzn2023.0.3
 harfbuzz                         x86_64                7.0.0-2.amzn2023.0.2
 java-17-amazon-corretto-headless x86_64                1:17.0.21.0.11-1.amzn2023.1
 java-21-amazon-corretto-headless x86_64                1:21.0.13.0.11-1.amzn2023.1
 javapackages-filesystem          noarch                6.0.0-7.amzn2023.0.6
 javapackages-tools               noarch                6.0.0-7.amzn2023.0.6
```

```
#!/bin/bash

SERVICE="tomcat"
NAMESPACE="Custom/Tomcat"

# Get EC2 Instance ID (IMDSv2)
TOKEN=$(curl -s -X PUT \
  http://169.254.169.254/latest/api/token \
  -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

INSTANCE_ID=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/instance-id)

REGION=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/placement/region)

# Check service status
if systemctl is-active --quiet "$SERVICE"; then
    STATUS=1
    echo "Tomcat is Running"
else
    STATUS=0
    echo "Tomcat is Stopped"
fi

# Send status to CloudWatch
aws cloudwatch put-metric-data \
  --namespace "$NAMESPACE" \
  --metric-name "TomcatServiceStatus" \
  --dimensions InstanceId="$INSTANCE_ID" \
  --value "$STATUS" \
  --unit Count \
  --region "$REGION"

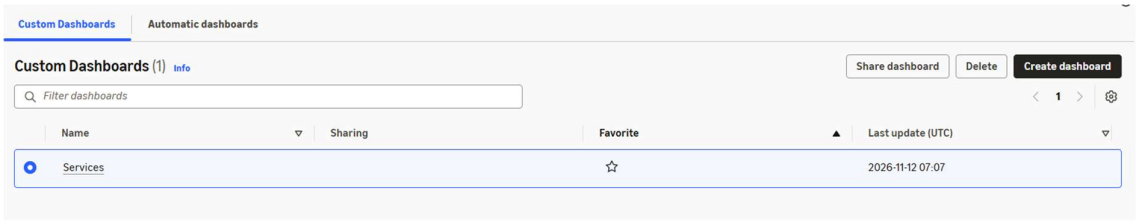
if [ $? -eq 0 ]; then
    echo "Service status sent to CloudWatch"
else
    echo "Failed to send service status"
fi
```

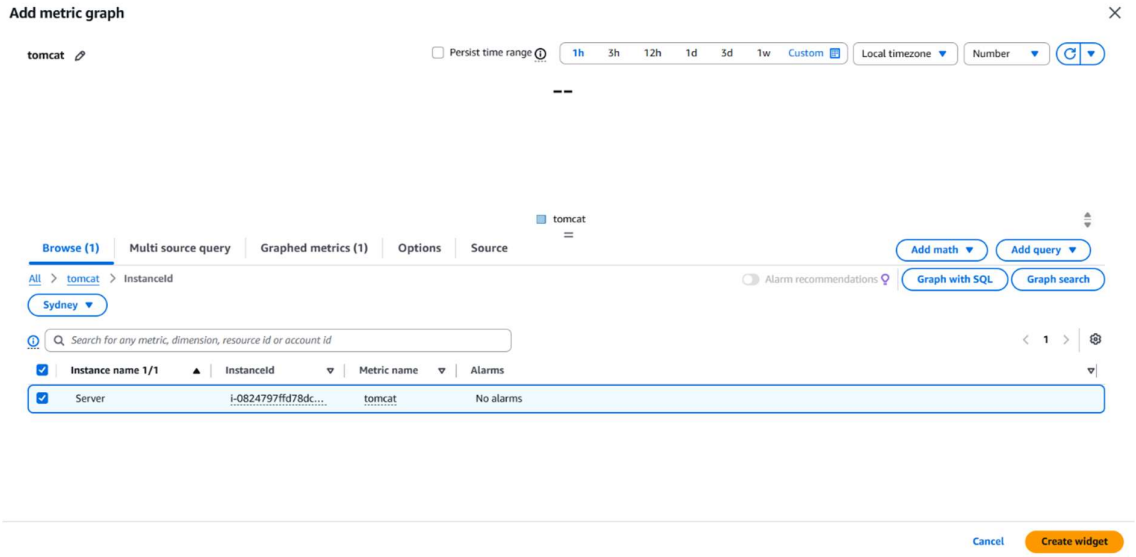
```
[root@ip-172-31-29-203 ~]# systemctl start tomcat10
[root@ip-172-31-29-203 ~]# systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-11-12 06:45:04 UTC; 56min ago
   Main PID: 26698 (java)
      Tasks: 29 (limit: 1059)
     Memory: 60.5M
        CPU: 7.049s
     CGroup: /system.slice/tomcat10.service
             └─26698 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.Basic

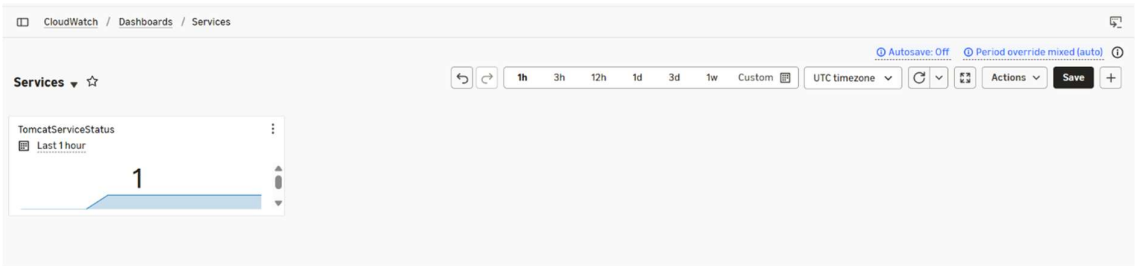
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.280 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.281 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.285 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.294 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.783 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.829 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.895 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.895 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.921 INFO [main] org.apach
Nov 12 06:45:05 ip-172-31-29-203.ec2.internal server[26698]: 12-Nov-2026 06:45:05.939 INFO [main] org.apach
lines 1-20/20 (END)
```

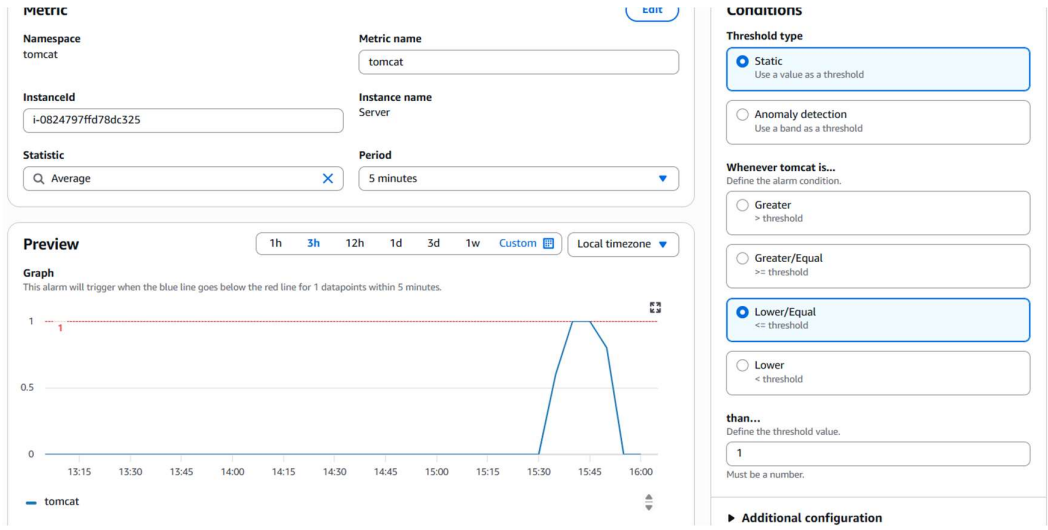
```
[root@ip-172-31-29-203 ~]# ./Managetomcat.sh
Tomcat is Running
Service status sent to CloudWatch
[root@ip-172-31-29-203 ~]#
```

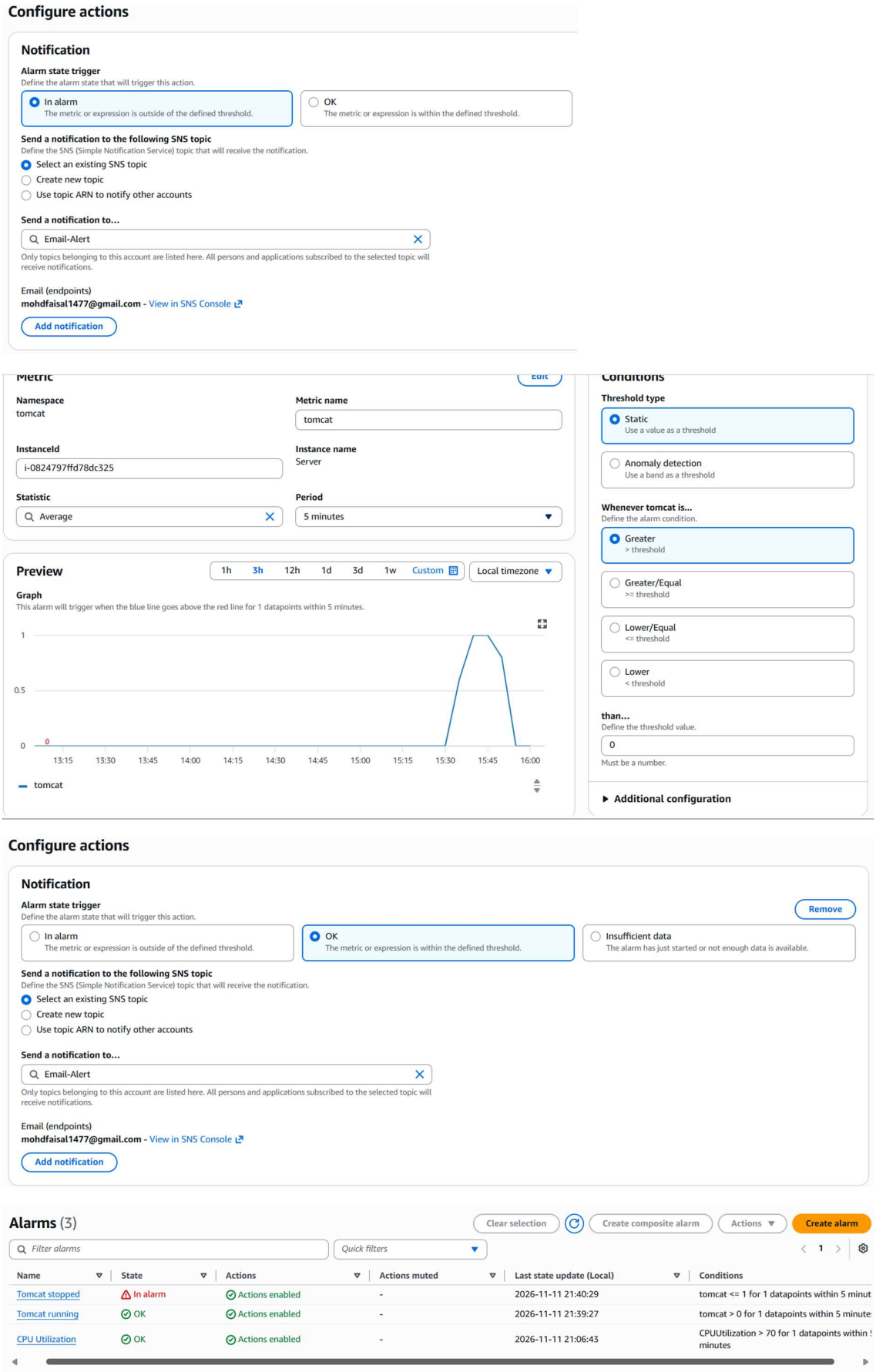
```
[root@ip-172-31-38-94 opt]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-172-31-38-94 opt]# crontab -l
* * * * * /opt/ManageTomcat.sh
```


Custom Dashboards
Automatic dashboards
Custom Dashboards (1) Info
Share dashboard
Delete
Create dashboard
Filter dashboards
Name
Sharing
Favorite
Last update (UTC)
Services
2026-11-12 07:07


Add metric graph
tomcat
Persist time range
1h 3h 12h 1d 3d 1w Custom
Local timezone
Number
tomcat
Browse (1)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
All > tomcat > InstanceId
Alarm recommendations
Graph with SQL
Graph search
Sydney
Search for any metric, dimension, resource id or account id
Instance name 1/1
InstanceId
Metric name
Alarms
Server
i-0824797ffd78dc...
tomcat
No alarms
Cancel
Create widget


CloudWatch / Dashboards / Services
Autosave: Off
Period override mixed (auto)
Services
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Actions
Save
TomcatServiceStatus
Last 1 hour
1


Metric
Edit
Namespace
tomcat
Metric name
tomcat
InstanceId
i-0824797ffd78dc325
Instance name
Server
Statistic
Average
Period
5 minutes
Preview
1h 3h 12h 1d 3d 1w Custom
Local timezone
Graph
This alarm will trigger when the blue line goes below the red line for 1 datapoints within 5 minutes.
1
0.5
0
13:15 13:30 13:45 14:00 14:15 14:30 14:45 15:00 15:15 15:30 15:45 16:00
tomcat
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever tomcat is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
1
Must be a number.
Additional configuration



6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Step 1:- First attach a role with permission cloudwatchagentserver-policy to the instance.

Step 2:-Then install the nignx package and run the package with systemctl.

Step 3:- Now create a bash script to monitor the tomcat status and send it to cloudwatch.

Step 4:-Change the permissions of the file and run the script.

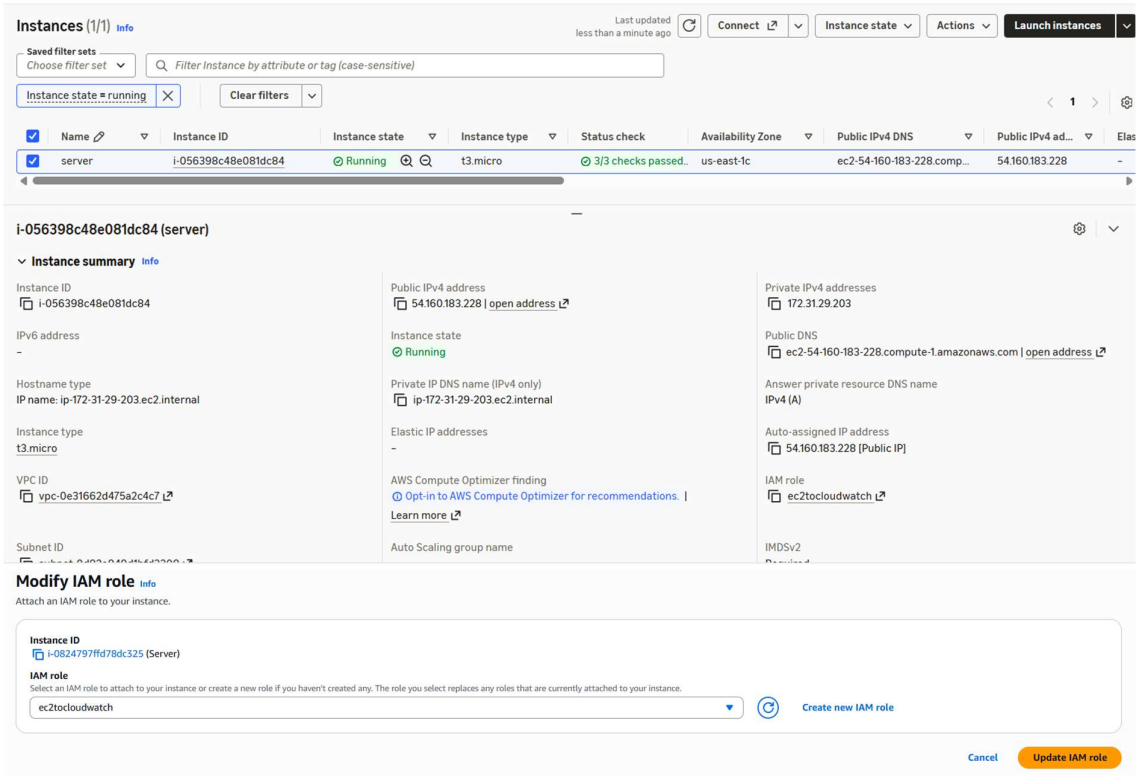
Step 5:- Now in cloudwatch dashboard when we click on new widget we can see new metric tomcat is added.

Step 6:- click the metric and add it to dashboard now edit the time to one minute.

Step 7:- To continuously monitor the service add the script to cronjobs.

Step 8:- Go to alarms to add new alarm set the condition such if the metric value is 1 status is ok.

Step 9:- If the status is 0 then the service file is not working. Create alarms for both conditions.



```
[root@ip-172-31-38-94 opt]# yum install nginx -y
Last metadata expiration check: 3:47:52 ago on Wed Nov 11 12:15:52 2026.
Dependencies resolved.
================================================================================
 Package                                 Architecture                 Version
================================================================================
Installing:
 nginx                                   x86_64                       1:1.30.6-1.amzn2023.0.1
Installing dependencies:
 generic-logos-httpd                     noarch                       18.0.0-12.amzn2023.0.3
 gperftools-libs                         x86_64                       2.9.1-1.amzn2023.0.3
 libunwind                               x86_64                       1.4.0-5.amzn2023.0.3
 nginx-core                              x86_64                       1:1.30.6-1.amzn2023.0.1
 nginx-filesystem                        noarch                       1:1.30.6-1.amzn2023.0.1
 nginx-mimetypes                         noarch                       2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(2/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm
(4/7): nginx-1.30.6-1.amzn2023.0.1.x86_64.rpm
(5/7): nginx-core-1.30.6-1.amzn2023.0.1.x86_64.rpm
(6/7): nginx-filesystem-1.30.6-1.amzn2023.0.1.noarch.rpm
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
```

```
[root@ip-172-31-38-94 opt]# systemctl start nginx
[root@ip-172-31-38-94 opt]# ./ManageNginx.sh
[root@ip-172-31-38-94 opt]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-11-11 16:03:59 UTC; 1min 23s ago
    Process: 19740 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 19741 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 19742 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 19743 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 57ms
     CGroup: /system.slice/nginx.service
             ├─19743 "nginx: master process /usr/sbin/nginx"
             ├─19744 "nginx: worker process"
             └─19745 "nginx: worker process"

Nov 11 16:03:59 ip-172-31-38-94.ap-southeast-2.compute.internal systemd[1]: Starting nginx.service - The ng
Nov 11 16:03:59 ip-172-31-38-94.ap-southeast-2.compute.internal nginx[19741]: nginx: the configuration file
Nov 11 16:03:59 ip-172-31-38-94.ap-southeast-2.compute.internal nginx[19741]: nginx: configuration file /et
Nov 11 16:03:59 ip-172-31-38-94.ap-southeast-2.compute.internal systemd[1]: Started nginx.service - The ngi
[root@ip-172-31-38-94 opt]#
```

```
#!/bin/bash

SERVICE="nginx"
NAMESPACE="Nginx"

# Get EC2 Instance ID (IMDSv2)
TOKEN=$(curl -s -X PUT \
  http://169.254.169.254/latest/api/token \
  -H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

INSTANCE_ID=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/instance-id)

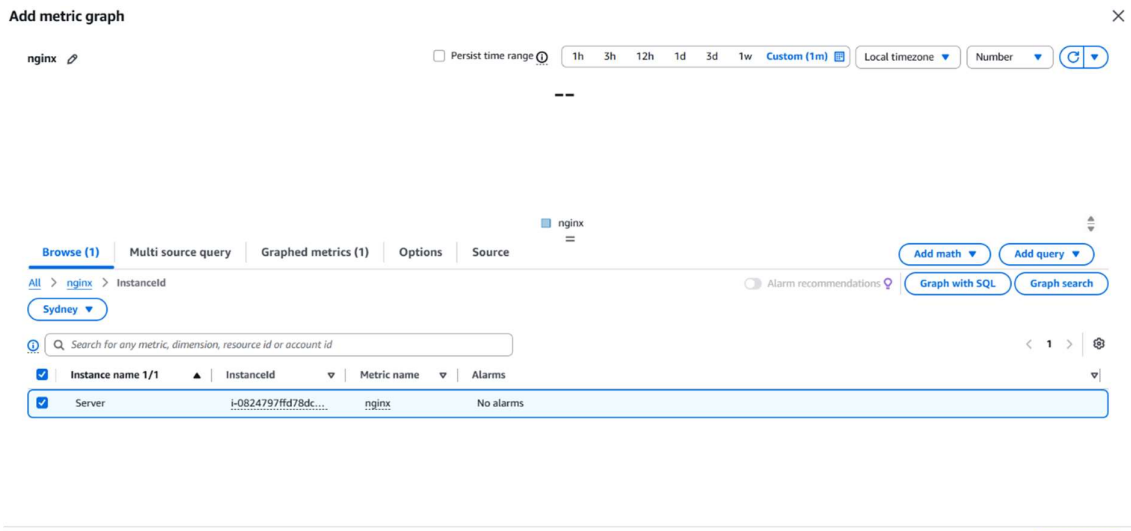
REGION=$(curl -s \
  -H "X-aws-ec2-metadata-token: $TOKEN" \
  http://169.254.169.254/latest/meta-data/placement/region)

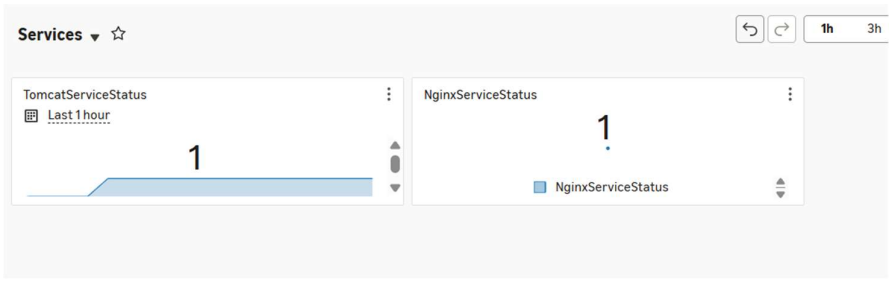
# Check service status
if systemctl is-active --quiet "$SERVICE"; then
    STATUS=1
    echo "Nginx is Running"
else
    STATUS=0
    echo "Nginx is Stopped"
fi

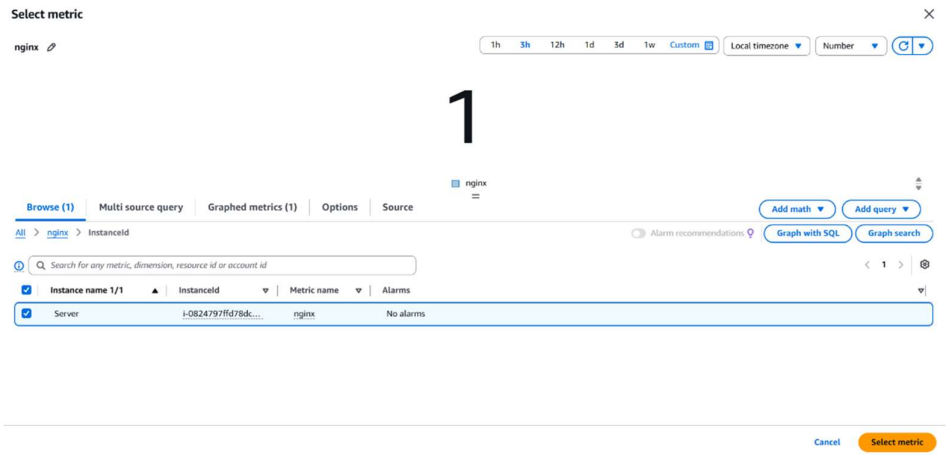
# Send status to CloudWatch
aws cloudwatch put-metric-data \
  --namespace "$NAMESPACE" \
  --metric-name "NginxServiceStatus" \
  --dimensions InstanceId="$INSTANCE_ID" \
  --value "$STATUS" \
  --unit Count \
  --region "$REGION"

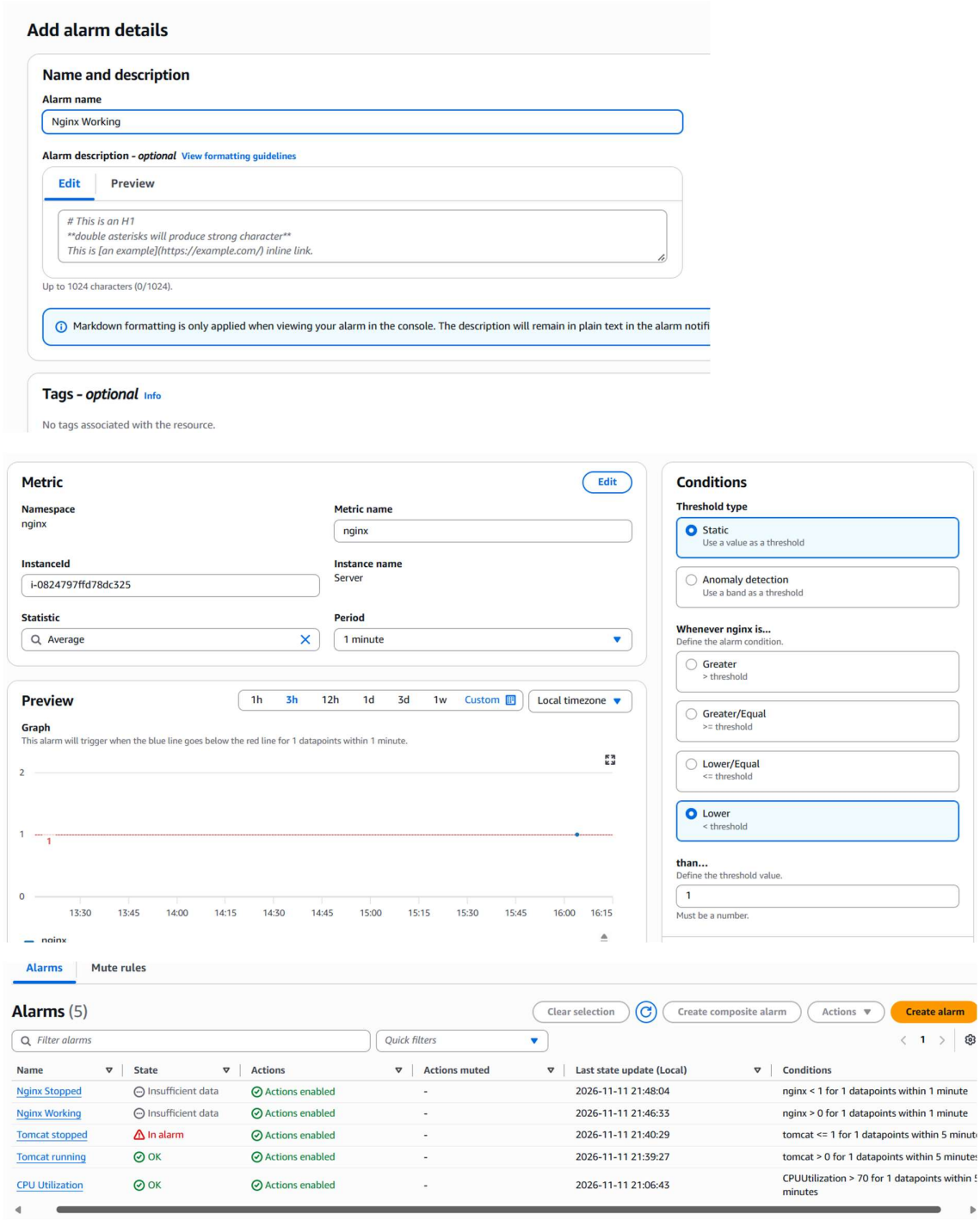
if [ $? -eq 0 ]; then
    echo "Service status sent to CloudWatch"
else
    echo "Failed to send service status"
fi
```

```
[root@ip-172-31-38-94 opt]# crontab -e
crontab: installing new crontab
[root@ip-172-31-38-94 opt]# crontab -l
* * * * * /opt/ManageTomcat.sh
* * * * * /opt/ManageNginx.sh
[root@ip-172-31-38-94 opt]#
```









Alarms | Mute rules

Alarms (5)

Clear selection | Create composite alarm | Actions | Create alarm

Filter alarms | Quick filters | 1

Name	State	Actions	Actions muted	Last state update (Local)	Conditions
Nginx Stopped	Insufficient data	Actions enabled	-	2026-11-11 21:48:04	nginx < 1 for 1 datapoints within 1 minute
Nginx Working	Insufficient data	Actions enabled	-	2026-11-11 21:46:33	nginx > 0 for 1 datapoints within 1 minute
Tomcat stopped	In alarm	Actions enabled	-	2026-11-11 21:40:29	tomcat <= 1 for 1 datapoints within 5 minut
Tomcat running	OK	Actions enabled	-	2026-11-11 21:39:27	tomcat > 0 for 1 datapoints within 5 minutes
CPU Utilization	OK	Actions enabled	-	2026-11-11 21:06:43	CPUUtilization > 70 for 1 datapoints within 5 minutes